

DESIGN ISSUES:

1. The CUSTOMER table uses an ID column but defines no primary key constraint on it, and the ORDERS table has no primary key at all, despite needing a composite one, both of which are critical data integrity failures.
2. The PRODUCT table defines UNIQUE (ID) where PRIMARY KEY (ID) should be used instead, which is an unusual and problematic design choice since a UNIQUE constraint permits multiple NULL values whereas a primary key does not.
3. Several columns that represent numeric identifiers have been incorrectly typed as VARCHAR2: OID in the ORDER table should be NUMBER, the CUSTOMER column in ORDER and the ORDER column in ORDERS should both be NUMBER foreign keys, and the SORT column in PRODUCT should also be NUMBER rather than VARCHAR2(25), using text types for ID fields fundamentally breaks relational integrity.
4. While the CUSTOMER.ZIP → PLACE.ZIP foreign key exists, four expected relationships are entirely absent: ORDER.CUSTOMER → CUSTOMER.ID, ORDERS.ORDER → ORDER.OID, ORDERS.PRODUCT → PRODUCT.ID, and a constraint linking PRODUCT.SORT to its lookup table, all of which are critical omissions.
5. Both ORDERS.ORDER and ORDERS.PRODUCT are defined as nullable when they should be NOT NULL, meaning the schema currently allows line items to exist with no associated order or product, which leads to orphaned records.
6. The ORDERS.AMOUNT column is defined as NUMBER(2,0), which caps the maximum orderable quantity at 99 items per line and should be widened to at least NUMBER(5,0) to be practically usable.
7. CUSTOMER.NO is typed as NUMBER(3,1), giving a street number a decimal place, and PRODUCT.SZ is NUMBER(3,2), allowing two decimal places for a size field, both are unusual choices, though they are technically valid.
8. PRODUCT.PRICE is defined as NUMBER(3,2), which restricts the maximum storable price to 9.99 and mirrors the same limitation seen in other branches of the schema.
9. While CUSTOMER.ID and PRODUCT.ID are correctly typed as NUMBER, ORDER.OID is VARCHAR2, creating an inconsistency in both naming convention and data type that introduces confusion and hinders relational joins.
10. PRODUCT.SORT is stored as VARCHAR2(25) when it should be a NUMBER foreign key referencing a SORT lookup table, meaning there is currently no relational integrity enforced on that column.
11. Every NOT NULL constraint in the schema has been duplicated as an additional CHECK constraint, when the NOT NULL declaration in the column definition alone is sufficient and the redundant checks should be removed.
12. It contains two tables ORDER and ORDERS that are almost similar and might lead to redundancy. Also ORDER is a reserved key.